
PROJECT PROPOSAL: MESH QUALITY IN MGNs AND BENCHMARKING SIMULATION METHODS OF PINNs, MGNs, AND FEM

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1 INTRODUCTION

This proposal briefly outlines my proposed project in the remaining weeks of the semester. It aims at addressing and understanding mesh quality while using Mesh Graph Networks (MGNs) as well as benchmarking simulation methods for solving more complex geometries. By comparing physics-informed neural networks (PINNs), MGNs, and traditional finite element methods (FEM), the project will assess each method's convergence behavior, accuracy, and computational efficiency. I hope to be able to find out the best way to generalize for different shapes, boundary conditions. I will also take a look at the very least why MGNs provide a jagged solution and potentially develop a suggestion or improvements to reduce jagged mesh artifacts in MGNs avoiding for loops and integrating these improvements into a broader simulation framework.

This project hopefully aims to (at least finish 2 points below):

- Identify and mitigate the causes of jagged meshes in MGNs.
- Benchmark PINNs, MGNs, and FEM on varying geometry, BCs, etc.
- Provide a comparative analysis that highlights the strengths and limitations of each approach in terms of accuracy, convergence, and computational performance.

2 METHODOLOGY

1. Problem Formulation and Test Case Development

- Benchmark Problems: Design test cases with varying geometries and different boundary conditions and loads by modifying and generalizing my current 2D Code.
- Unified Simulation Framework: Develop a MATLAB environment that allows for implementation and comparison of PINNs, MGNs, and FEM.

2. Understanding MGNs

- Graph Smoothing Techniques: Implement algorithms that adjust node connectivity or edge weights to smooth out irregularities maybe look into adaptive node re-distribution,
- Investigate the current graph-based representation in MGNs to identify the origins of jagged solutions.

3. Benchmarking Simulation Methods

- Implement PINNs and FEM models for the defined benchmark problems, ensuring consistency in boundary conditions and problem setup.
- Establish quantitative metrics such as: Convergence rate to achieve a specific tolerance. Numerical error via the ℓ_2 norm. Computational cost (runtime and memory usage)

4. Data Analysis and Comparative Evaluation

- Interpret results to draw conclusions about the strengths and limitations of each method and recommend areas for further improvement.
- Present results through plots and tables that highlight convergence behavior, error trends, and computational efficiency.

3 IMPLEMENTATION TIMELINE

Week 1: Literature Review and Planning — Review MGN research and check for pre-existing benchmark cases, and set up the MATLAB simulation framework. **Week 2:** Analyze current MGN issues; implement graph smoothing if possible or suggest alternatives to for loops. **Week 3-4:** Develop PINNs and FEM models; integrate improved MGNs into the framework. **Week 5:** Execute simulations; collect performance data and perform initial analysis. **Week 6:** Complete comparative evaluation and compile results into a report and presentation.

4 FINAL GOALS & EVALUATION

By the end of the semester, I aim to have understood the enhanced MGN implementation that minimizes jagged artifacts and a simulation framework that benchmarks PINNs, improved MGNs, and FEM. Our goal is to demonstrate measurable improvements or understanding of convergence, accuracy, and computational efficiency when solving complex geometries using PINNs, MGNs, and FEM.

- **Decomposition into Sub-Goals:** Our final goal is decomposed into the following sub-goals:
 - **Result A (Achievable):** Benchmark MGN with PINNs and FEM in the simulation framework and demonstrate which has the superior performance and which is computationally efficient and generalizable.
 - **Result B (Ambitious):** Understand why MGNs provide jagged solutions and implement a basic version of the enhanced MGN with graph smoothing and adaptive node re-distribution.

Our immediate objective is to achieve **Result A** and report its success as it is fundamental to the project, while we work towards **Result B** as the ambitious, long-term goal.